

IDS2Eval Scorecard

cic-ids2017

12 ok · 4 warning · 3 flag · 19 checks

REVIEW REQUIRED

- **dedup_check**
train duplicates: 0 (0.00%); test rows leaking a train feature-match: 0 (0.00%); test-internal duplicates: 0

ok
- **label_conflict_check**
no feature vector maps to more than one label

ok
- **near_duplicate_class_check**
2 of 6,029 sampled rows (0.03%) have a near-zero-distance neighbor under a different label. Most affected pairs: {'DoS Slowhttptest / DoS slowloris': 2}

flag
- **leakage_screen**
no single feature or pair dominates importance (top1=6.7%, top2=12.7%)

ok
- **one_rule_check**
single rule 'Bwd Packet Length Std <= 1495' reaches 90.0% test accuracy (89.0% train), vs. 83.7% from always predicting the majority class

ok
- **feature_auc_ranking_check**
'Bwd Packet Length Max' alone reaches AUC 1.000 identifying 'Heartbleed'; a single feature this separable is worth checking for a leakage artifact

flag
- **identity_column_flag**
standalone AUC by column (seen-value coverage, AUC on seen values): Destination Port: 0.936 (99.4%, 0.935). Suggest dropping: ['Destination Port']

flag
- **port_protocol_shortcut_check**
no port-like entry in schema.id_like_columns and/or no proto-like column found, this check needs both

ok
- **temporal_leakage_check**
no schema.timestamp_column configured

ok
- **temporal_realism_check**
no schema.timestamp_column configured

ok
- **flow_group_leakage_check**
no schema.flow_id_columns configured

ok
- **row_order_leakage_check**
adjacent-row label-transition rate (train=1.00, test=1.00) as a share of what a randomly shuffled ordering would produce

ok
- **homogeneity_test**
12 classes tested; for every class the 95% CI upper bound on (test-to-train minus train-internal) near-duplicate rate is below the +0.05 margin, so no class shows a train/test leakage signature

ok
- **scenario_holdout_falsification**
no schema.scenario_column configured

ok
- **class_distribution_report**
15 classes, train imbalance ratio (majority:minority) = 185530:1; classes below 1% of train: ['DoS GoldenEye', 'FTP-Patator', 'DoS slowloris', 'DoS Slowhttptest', 'SSH-Patator', 'Bot', 'Web Attack ⚡ Brute Force', 'Web Attack ⚡ XSS', 'Infiltration', 'Web Attack ⚡ Sql Injection', 'Heartbleed']

warning
- **low_cardinality_warning**
unique values by column: {'Destination Port': 51291}

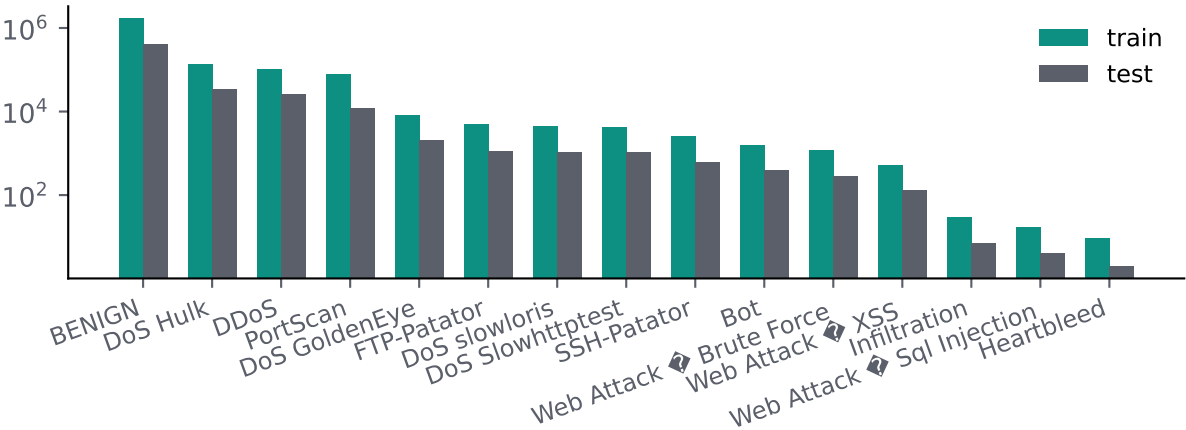
ok
- **schema_fingerprint_check**
matched extractor signature(s): ['CICFlowMeter']

warning
- **data_integrity_check**
missing values in 1 feature column(s); +-inf values in 2 feature column(s); 8 constant/near-constant feature(s)

warning
- **known_issue_lookup**
The traffic capture has packet misorder and duplication, and some attacks that were actually launched are not correctly labeled as attack traffic in the released CSVs. (Engelen et al. 2021, IEEE S&P Workshops (WTMC)); A full re-labeling of the original release found 6.67% of labels wrong, with the error rate above 75% for some attack classes. (Liu et al. 2022, IEEE CNS (rates as summarized by Cantone et al. 2024)); Heartbleed has only 11 rows (about 0.0004% of the dataset), too few to evaluate reliably regardless of labeling accuracy. (Cantone et al. 2024, IEEE Access, Table 1)

warning

Class distribution (log scale)



Top feature importances (leakage_screen)

